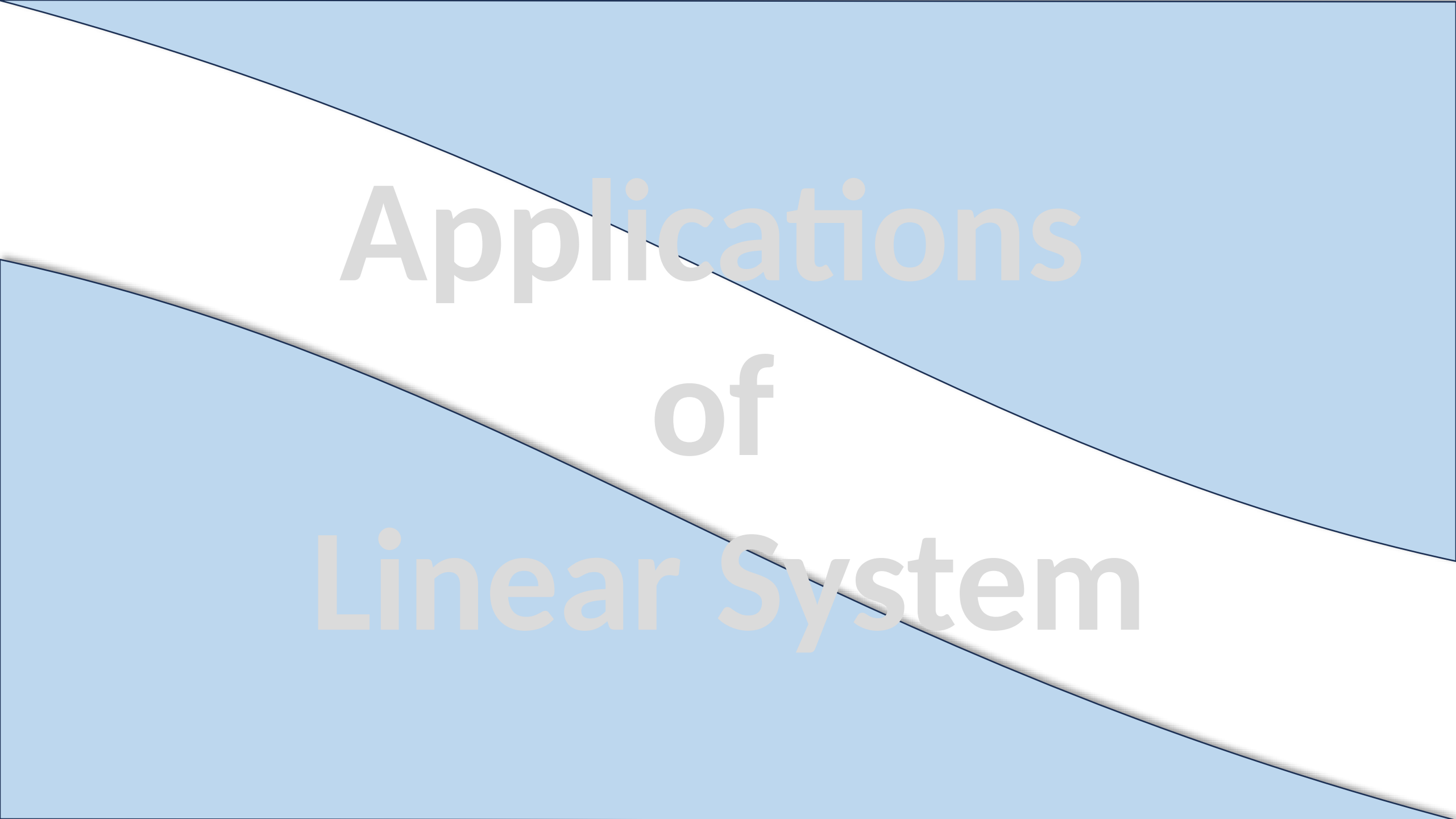




Applications of Linear System

Real-World Problem: Allocating Computing Resources in a Data Center

A **cloud computing company** provides virtual machines (VMs) to different clients.

The company has **three types of computing resources**:

- **CPU cores**
- **RAM in GB**
- **Storage in TB**

Each **client request** requires a specific combination of these resources. The data center must allocate resources efficiently while ensuring **all resources are fully utilized**.

$x =$

Problem Setup: Resource Allocation as a System of Linear Equations

The company receives **three client requests**, and the total available resources are:

• **60 CPU cores** . **130 GB RAM** . **100 TB Storage**

Each client's request results in a system of three equations:

x represents the **number of times Client 1's request** is fulfilled.

y represents the **number of times Client 2's request** is fulfilled.

z represents the **number of times Client 3's request** is fulfilled.

1 Client 1: Requests 10 CPU cores, 20 GB RAM, and 30 TB Storage

2 Client 2: Requests 10 CPU core, 10 GB RAM, and 20 TB Storage

3 Client 3: Requests 10 CPU cores, 30 GB RAM, and 10 TB Storage

$$10x + 10y + 10z = 60$$

$$20x + 10y + 30z = 130$$

$$30x + 20y + 10z = 100$$

We need to solve for (x, y, z) using the **Gauss-Jordan Method** to determine how much each resource is allocated per request.

Solution using Gauss-Jordan Elimination Method

We represent the system as an **augmented matrix**:

$$\left[\begin{array}{ccc|c} 10 & 10 & 10 & 60 \\ 20 & 10 & 30 & 130 \\ 30 & 20 & 10 & 100 \end{array} \right]$$

Solving for **x , y , and z** , we get:

$$x = 1, \quad y = 2, \quad z = 3$$

2. Server Load Balancing Optimization

A cloud computing company manages a **server load balancing system**, where incoming requests are distributed among **three servers** (S_1 , S_2 , S_3). The company wants to ensure **balanced utilization** of these servers based on processing time.

The total **processing time** (in milliseconds) for the requests on each server is governed by the following three **linear equations**:

$$x + y + z = 100$$

$$2x + 3y + 4z = 300$$

$$x + 2y + 3z = 200$$

where:

- x = Requests assigned to **Server 1** (S_1)
- y = Requests assigned to **Server 2** (S_2)
- z = Requests assigned to **Server 3** (S_3)

How This System Is Generated

This system models **server load balancing** in a cloud computing scenario, where x , y , and z represent the processing time (in milliseconds) for three servers S_1 , S_2 , and S_3 , respectively. Each equation reflects certain constraints or rules imposed by the system administrators to distribute the load in an optimal way.

$$x + y + z = 100$$

This represents the **total load constraint** across all servers.

Example interpretation: The total processing time of incoming tasks distributed to the servers must sum up to **100 milliseconds**.

Solution using Gauss-Jordan Elimination Method

We represent the system as an **augmented matrix**:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 100 \\ 2 & 3 & 4 & 300 \\ 1 & 2 & 3 & 200 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 100 \\ 0 & 1 & 2 & 100 \\ 0 & 1 & 2 & 100 \end{array} \right] \quad \begin{array}{l} R_2 - 2 * R_1 \\ R_3 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 100 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} R_1 - R_2 \\ R_3 - R_2 \end{array}$$

Rewriting as equations:

$$x - z = 0 \Rightarrow x = z$$

$$y + 2z = 100 \Rightarrow y = 100 - 2z$$

Let $z = t$ (since it can take any real value).

Thus the solution is:

$$x = t$$

$$y = 100 - 2t$$

$$z = t$$

Since z can take any real value, there are **infinitely many solutions**, meaning multiple ways to distribute requests across the servers.

Real-World Scenario for No Solution Case (Inconsistent System)

A **drone delivery company** is managing flight routes for three drones (D_1 , D_2 , and D_3) that deliver packages across three cities (City A, City B, and City C). Due to specific fuel, time, and distance constraints, they try to solve a set of conditions that the drones can make per day.

The company sets the following constraints based on drone operational limits:

- 1 The total number of trips made by all drones must equal **50 trips per day**:

$$x + y + z = 50$$

- 2 Due to fuel limitations, the combined total of trips by Drone 1 and Drone 2 must be **60 trips**:

$$x + y = 60$$

- 3 A delivery quota requires Drones 2 and 3 to make a total of **30 trips**:

$$y + z = 30$$

Where:

x = Number of trips by **Drone 1 (D_1)**

y = Number of trips by **Drone 2 (D_2)**

z = Number of trips by **Drone 3 (D_3)**

Solution using Gauss-Jordan Elimination Method

We represent the system as an **augmented matrix**:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 1 & 1 & 0 & 60 \\ 0 & 1 & 1 & 30 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & 0 & -1 & 10 \\ 0 & 1 & 1 & 30 \end{array} \right] \quad \boxed{R_2 - R_1}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & 1 & 1 & 30 \\ 0 & 0 & -1 & 10 \end{array} \right] \quad \boxed{R_3 \leftrightarrow R_2}$$

Now, the system gives $z = -10$

However, in the context of drone trips, **negative values don't make sense** because z represents the number of trips by Drone 3. You can't have **-10 trips**!

This is a clear **real-world inconsistency**. The system of equations has **no solution**, which means the constraints are contradictory.

In practical terms, this inconsistency could indicate:

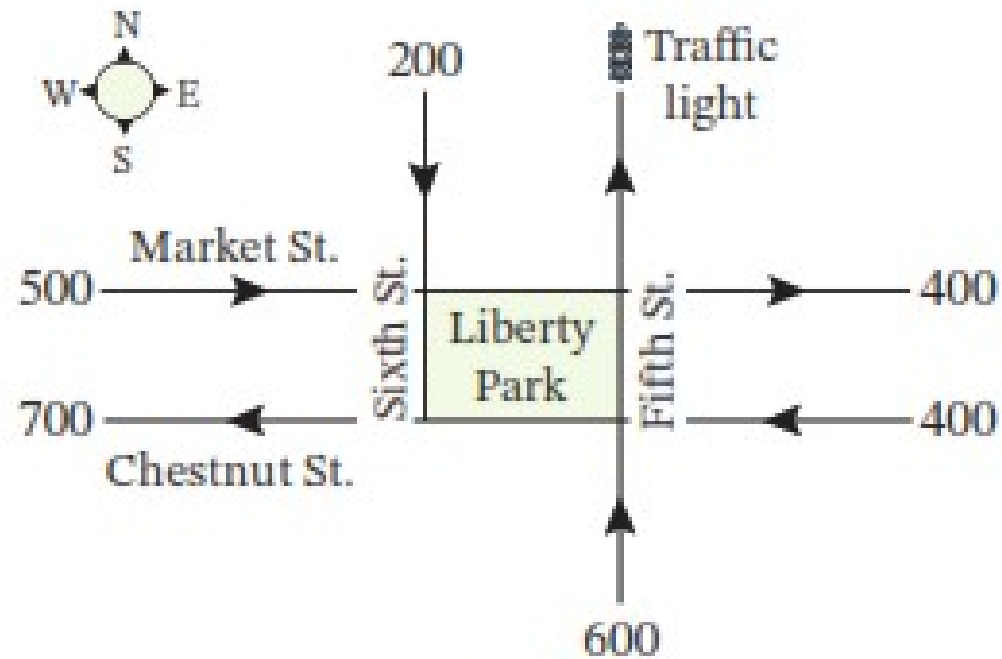
- 1.Conflicting Operational Constraints:** The company may have set impossible rules, like assigning more trips than the total capacity.
- 2.Resource Allocation Error:** Trying to distribute trips in a way that exceeds drone capabilities.
- 3.Error in Planning:** The constraints don't align with the reality of the system, and adjustments are needed.

Design of Traffic Patterns

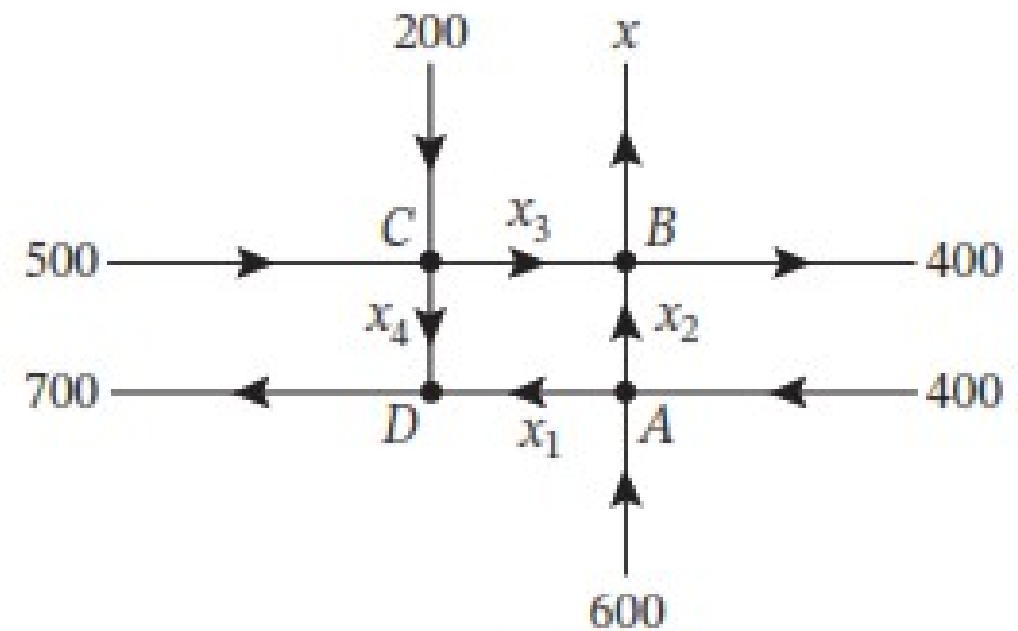
The network in Figure (next slide) shows a proposed plan for the traffic flow around a new park that will house the Liberty Bell in Philadelphia, Pennsylvania. The plan calls for a computerized traffic light at the north exit on Fifth Street, and the diagram indicates the average number of vehicles per hour that are expected to flow in and out of the streets that border the complex. All streets are one-way.

(a) How many vehicles per hour should the traffic light let through to ensure that the average number of vehicles per hour flowing into the complex is the same as the average number of vehicles flowing out?

(b) Assuming that the traffic light has been set to balance the total flow in and out of the complex, what can you say about the average number of vehicles per hour that will flow along the streets that border the complex?



(a)



(b)

Solution (a) If, as indicated in Figure 1.10.3b, we let x denote the number of vehicles per hour that the traffic light must let through, then the total number of vehicles per hour that flow in and out of the complex will be

Flowing in: $500 + 400 + 600 + 200 = 1700$

Flowing out: $x + 700 + 400$

Equating the flows in and out shows that the traffic light should let $x = 600$ vehicles per hour pass through.

Solution (b) To avoid traffic congestion, the flow in must equal the flow out at each intersection. For this to happen, the following conditions must be satisfied:

Intersection	Flow In		Flow Out
A	$400 + 600$	$=$	$x_1 + x_2$
B	$x_2 + x_3$	$=$	$400 + x$
C	$500 + 200$	$=$	$x_3 + x_4$
D	$x_1 + x_4$	$=$	700

Thus, with $x = 600$, as computed in part (a), we obtain the following linear system:

$$x_1 + x_2 = 1000$$

$$x_2 + x_3 = 1000$$

$$x_3 + x_4 = 700$$

$$x_1 + x_4 = 700$$

By Gauss-Jordan Method

We have

$$x_1 = 700 - t, \quad x_2 = 300 + t, \quad x_3 = 700 - t, \quad x_4 = t$$

However, the parameter t is not completely arbitrary here, since there are physical constraints to be considered.

So, we have

$$0 \leq x_1 \leq 700$$

$$300 \leq x_2 \leq 1000$$

$$0 \leq x_3 \leq 700$$

$$0 \leq x_4 \leq 700$$

THANKS